

COMP1008: An Overview of Part III (Intelligent Systems)

- ▶ Last week
 - Bayes Rule, KRR
 - Feedback: exercises
 - Update: coursework released
- ▶ This week
 - Natural Language Processing
 - ChatGPT (GenAI)
 - Feedback: exercises, EMF
- ▶ Following weeks: Part IV
 - Search techniques
 - Game playing

COMP1008 - Fundamentals of AI

Natural Language Processing (NLP)

Professor Rong Qu
rong.qu@nottingham.ac.uk

		text	spam
0	Subject: naturally irresistible your corporate...		1
1	Subject: the stock trading gunslinger fanny i...		1
2	Subject: unbelievable new homes made easy im ...		1
3	Subject: 4 color printing special request add...		1
4	Subject: do not have money , get software cds ...		1
5	Subject: great nnews hello , welcome to medzo...		1
6	Subject: here ' s a hot play in motion homela...		1
7	Subject: save your money buy getting this thin...		1
8	Subject: undeliverable : home based business f...		1
9	Subject: save your money buy getting this thin...		1



Milestones in AI

- ▶ 1997 **Deep Blue**
 - 11 May 1997, AI Game milestone
- ▶ 2011 **IBM Watson**
 - Won US quiz show Jeopardy
- ▶ 2014 **Turing test** “passed”
 - Chatbot Eugene Goostman
- ▶ 2016 **AlphaGo**
 - All open board games conquered
- ▶ 2022 **ChatGPT**
 - Strong AI (AGI)?

AI	COMP1008	CS Courses
Mixed	KRR + Tree Search	COMP1008 COMP3008 – Symbolic AI
Mixed	NLP + Machine learning + KRR	COMP1008; COMP3074 COMP3009 – MLE
Mixed	NLP, KRR	COMP3074 – Human AI Interaction; COMP3008
Mixed	Alpha-beta Search + Machine learning	COMP1008 COMP3009 – MLE
Mixed	Machine learning + NLP	COMP3009 – MLE COMP3074

Natural Language Processing (NLP)

- ▶ Human communicate using natural languages
 - Text: books, tweet, emails, etc.
 - Speech: audio, video
- ▶ Machines communicate
 - High-level programming languages, e.g. Python, Java, C
 - '0's and '1's
- ▶ **NLP**
 - Branch of AI which allows machines to “understand” and generate human languages



NLP – Approaches

- a) **Symbolic**: rely on rules and logic to represent and process languages
- Rule based: defines a set of instructions
 - Grammar based: linguistic grammars to analyse words interaction in sentences
 - Lexical databases: store words definitions and their relationships in context

COMP1008	CS Course
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KRR	COMP3008 – Symbolic AI
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- b) **Statistics**
- Probabilistic models: analyse language patterns to predict words, e.g. Bayes nets, etc.
 - N-grams: analyse sequences of n consecutive words to predict the next word

COMP1008	CS Course
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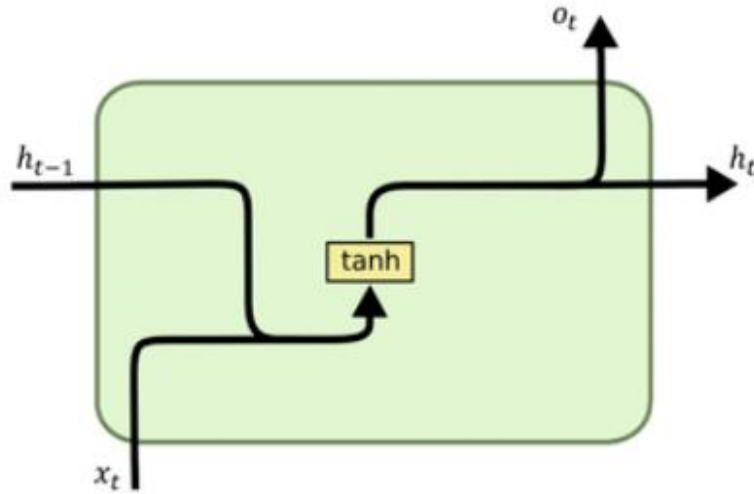
NLP	COMP3074 – Human AI Interaction
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- ▶ Still in use: easy to understand, reliable

NLP – Approaches

c) ANN based

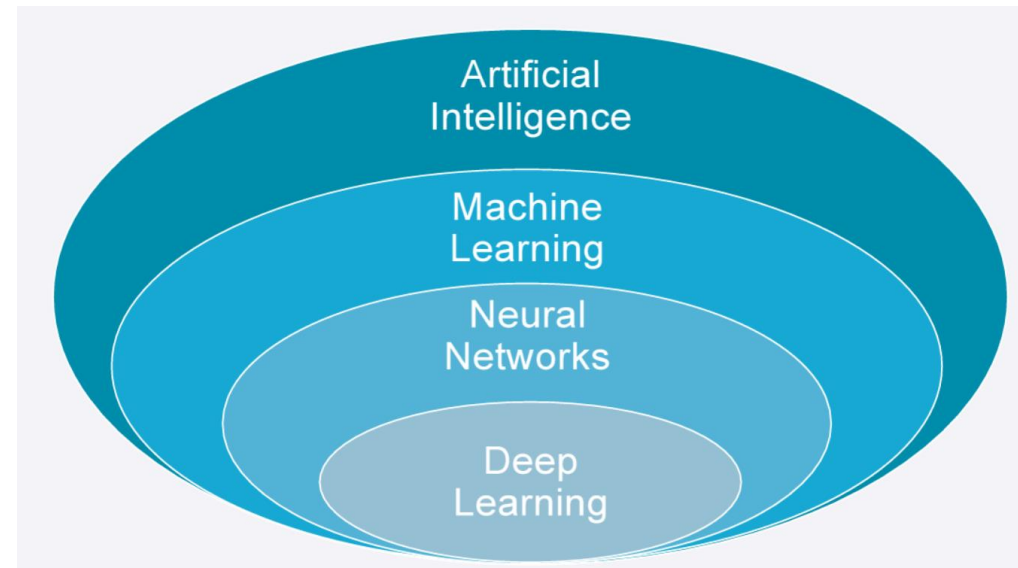
- Deep learning – Recurrent Neural Networks (RNN), Transformers
- Handle **sequential** data (e.g. text, speech, time series data) with **memory** (previous context)
- NLP revived - popular applications with the current machine learning development!



(a) Simple RNN

Machine Learning

COMP3009



NLP – Pipeline

COMP1008 COMP3074

NLP

Human AI Interaction

- ▶ A series of steps to transform **raw text** into a **format** suitable for further analysis and extract meaningful information

Text pre-processing

Normalisation

Tokenisation

Stopword
removal

Stemming

'Hey wanna win a £10k car? Win a Tesla!'

Feature engineering

Part-of-Speech
(POS) Tagging

Named Entity
Recognition (NER)

Advanced processing (applications)

Sentimental
analysis

Auto
correction

Machine
translation

Question
answering

Text
generation

Text
classification

Plagiarism
checker

ChatBots

Many many
more!!!

NLP – Pipeline

1. Text pre-processing

- **Normalization**: Convert text to standard form, e.g. lowercase, no punctuations, etc.
- **Tokenization**: Break down the text into words or phrases
- **Stopword Removal**: Filter out common words, e.g. "the", "is", "and"
- **Stemming**: Reduce words to their root form, e.g. 'running' to 'run'

'Hey wanna win a £10k car? Win a Tesla!'

hey	wanna	win	a	10k	car	Tesla
hey	wanna	win		10k	car	Tesla
hi	want	win		10000	car	Tesla

Text pre-processing

Normalisation

Tokenisation

Stopword
removal

Stemming

NLP – Pipeline

2. Feature engineering

- **Part-of-Speech (POS) Tagging**: assign each word its function, i.e. noun, verb, adjective, etc.
- **Named Entity Recognition (NER)**: identify named **entities** such as person, number, organisation, location and object in the text

'Hey wanna win a £10k car? Win a Tesla!'

			organisation		
			number	object	/ person?
exclamation	verb	verb	number	noun	noun
hi	want	win	10000	car	Tesla

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Feature engineering

Part-of-Speech
(POS) Tagging

Named Entity
Recognition (NER)

NLP – Pipeline

3. Applications / advanced processing

- a) **Symbolic based**: capitalisation/contextual rules (e.g. if first letter in upper case - then ..., if followed by Inc. - then ...)
- b) **Statistical models**: statistical models (e.g. **Bayes**) calculate the probability of the tags (e.g. surrounding words) or weights of features (e.g. capitalisation)
- c) **ANN based**: deep learning (e.g. RNN, transformers) to capture the semantic relationship in sequences of words

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					organisation
				number	object / person?
exclamation	verb	verb		number	noun noun
hi	want	win		10000	car Tesla

Advanced processing (applications)

Sentimental analysis

Auto correction

Machine translation

Question answering

Text generation

Text classification

Plagiarism checker

ChatBots

Many many more!!!

NLP – Milestones

- ▶ 2014 **Turing test** passed: Chatbot Eugene Goostman
 - **Rule-based systems**
 - Pre-determined responses / actions to prompts with **Statistical** models
 - Large database of pre-written responses and conversational **templates**
 - **Deception techniques**
 - Human and storytelling :-)
 - Pretend to be a child
 - Lack of transparency about the algorithms



AI	COMP1008	CS Courses
Mixed	NLP, KRR	COMP3074 – Human AI Interaction COMP3008 – Symbolic AI

NLP – Milestones

► 2011 IBM Watson

◦ NLP

- **POS tagging, NER**
- Syntactic analysis (grammar), Semantic analysis (sentiment, concepts, etc.)

◦ Machine learning

- **Supervised learning:** classification (categories), regression (predict values)
- **Un-supervised learning:** group similar topics based on patterns
- **Deep learning:** RNNs to learn complex patterns / sequences

◦ Reasoning and inference

- **Rule-based system:** predicate logic
- **Probabilistic reasoning:** calculate the probabilities of certain outcomes

Feature engineering

Part-of-Speech
(POS) Tagging

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AI

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CS Courses

Mixed

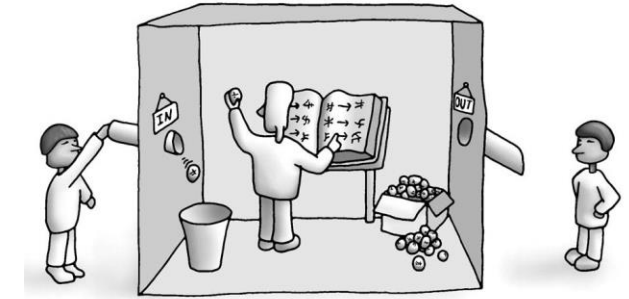
NLP + Machine learning + KRR

COMP3074; COMP3008; COMP3009

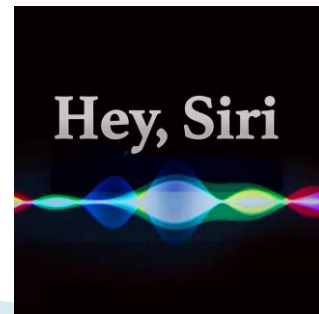
NLP – Applications

► Google Translate

- 2006-2016: Used to use traditional **statistical** methods
- Since 2017: **Neural Machine Translation** (ANNs)
 - Encoder: analyse and transfer words to useful representation
 - Decoder: generate words by word based on context
 - **Attention mechanism**: focus on specific parts of the sentence
 - Doesn't disclose all specific algorithms



Question: on your mobile, how many apps use NLP?



best wishes,

...

Yes, that is correct.

Yes to both questions.

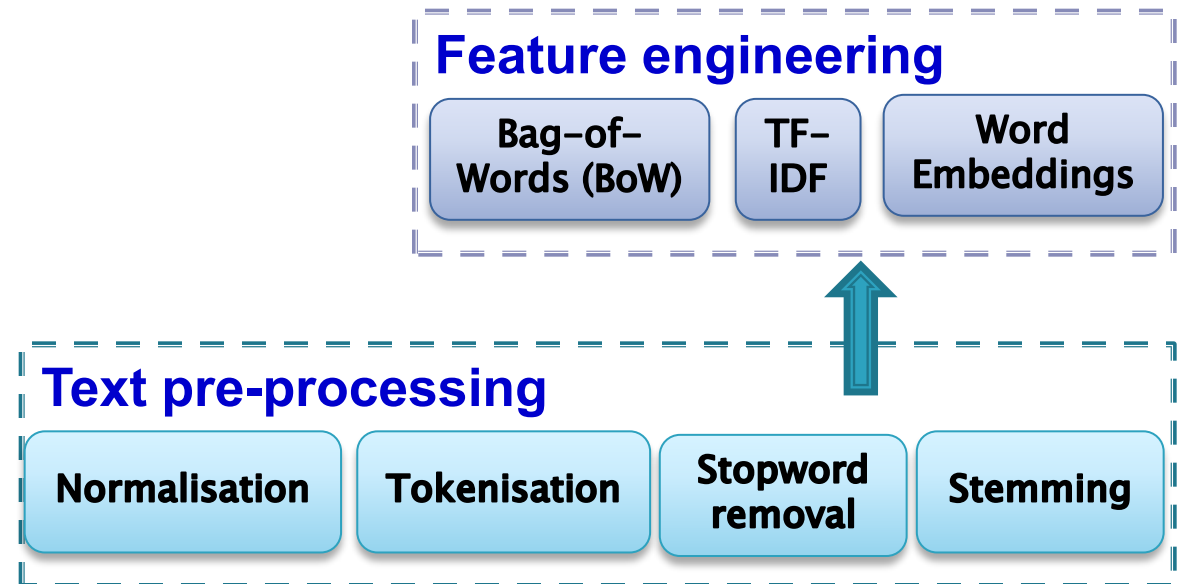
Yes.



NLP – Example: Spam emails classification

- ▶ **Text pre-processing**: obtain the **vocabulary** from the corpus, i.e. all documents
- ▶ **Feature engineering**
 - Bag-of-Words (**BoW**): format text data into numerical **matrix** (**rows**: documents, **columns**: vocabulary words, **cells**: frequencies of a word)
 - Term Frequency-Inverse Document Frequency (**TF-IDF**): calculate **importance** of words in a document relative to their frequency
 - **Word Embeddings**: transform into **vectors** of continuous values

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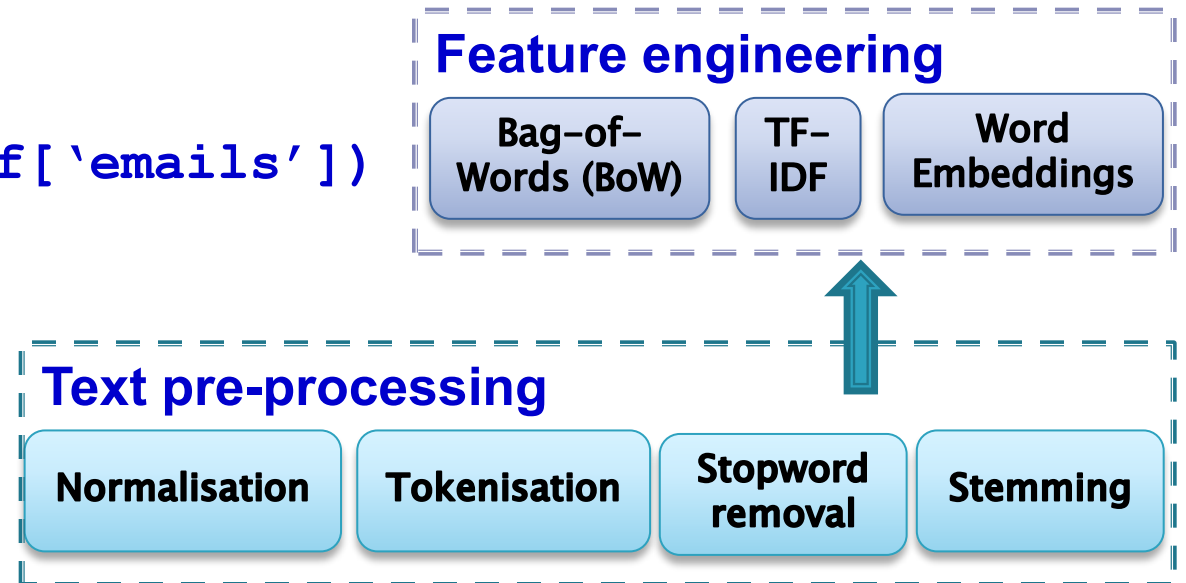


NLP – Example: Spam emails classification

- ▶ Text pre-processing: obtain the **vocabulary**
- ▶ Feature engineering: format text data to BoW numerical **matrix** (**rows**: documents, **columns**: vocabulary words, **cells**: frequencies of a word); TF-IDF: calculate **importance** of words; transform words into **vectors** of continuous values

```
from sklearn.feature_extraction.text import TfidfVectorizer  
# Convert a dataset of raw documents to a matrix of TF-IDF features
```

```
tf_vec = TfidfVectorizer()  
df_input = tf_vec.fit_transform(df['emails'])  
df_output = df['class']
```



NLP – Example: Spam emails classification

```
X_train,X_test,y_train,y_test = train_test_split(df_input,df_output)
from sklearn.naive_bayes import MultinomialNB
model = MultinomialNB()
model.fit(X_train, y_train)

predicted = model.predict(X_test)

msg = ['Hey wanna win a £10k car? Win a Tesla!'] # test an email
pred = model.predict(msg) # test on the test dataset
```

